

Latent Resonance: Zero-Shot Autoencoder Inversion and Azimuthal Spectral Forensics for Diffusion Image Attribution

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Abstract

Modern latent diffusion models (LDMs) synthesize photographic imagery with fidelity that evades human perception and frustrates conventional convolutional neural network classifiers. Existing forensic detectors rely predominantly on supervised binary classifiers that overfit to specific semantic domains, collapse under common post-processing transforms, or depend on closed API parameters. This paper introduces **Latent Resonance**, a zero-shot, training-free forensic framework grounded in the physical and architectural asymmetries between optical camera acquisition and latent autoencoder synthesis. By mapping candidate images through a deterministic autoencoder projection $x_{\square} = D(\mu(E(x)))$ without noise injection ($\sigma = 0$), we isolate three distinct physical and spectral phenomena: spatial manifold resonance, transposed-convolution harmonic lattice spikes, and CMOS Photo-Response Non-Uniformity (PRNU) photodiode noise independence.

Synthetic diffusion images originate directly from the decoder manifold $D(z)$, yielding near-zero spatial reconstruction error ($\text{PSNR} = 37.15 \pm 0.60$ dB). Conversely, authentic optical photographs suffer irreversible loss of physical sensor shot noise and PRNU across the $8\times$ latent spatial bottleneck, producing significantly higher residual error ($\text{PSNR} = 33.28 \pm 0.96$ dB, $\Delta = +3.88$ dB, Cohen's $d = 4.84$, $p = 2.88 \times 10^{-10}$). In the frequency domain, azimuthally averaged 2D Fast Fourier Transform (2D-FFT) analysis reveals that diffusion residuals exhibit sharp periodic harmonic spikes ($3.655\times$ baseline) induced by transposed convolution upsampling strides, whereas authentic photos follow a smooth, continuous $1/f^\alpha$ power-law decay ($1.144\times$). Furthermore, high-frequency inter-channel Laplacian correlation demonstrates near-perfect cross-color coupling in synthetic neural tensors ($\rho_{\text{RGB}} = 0.981 \pm 0.011$) compared to independent physical silicon photodiodes in optical cameras ($\rho_{\text{RGB}} = 0.000 \pm 0.001$). On a comprehensive large-scale benchmark of $N=1,000$ verified images (500 authentic camera captures, 500 generative diffusion outputs across SDXL and SD 1.5), Latent Resonance achieves an empirical **100.00% AUROC** with **0.00% False Accusation Rate (FAR)** and 100.00% True Positive Rate (TPR). Under adversarial stress testing (lossy JPEG $Q \in \{95, 85, 75\}$, bicubic resampling $384 \rightarrow 512$, and Gaussian blur $\sigma = 0.8$), the multi-signal triad provides robust forensic discrimination where single-domain detectors collapse. All benchmark datasets, raw prediction records, and interactive inspection interfaces are released openly for reproducible peer review.

1. Introduction

The proliferation of high-capacity latent diffusion models (LDMs) has dismantled traditional visual watermarks of synthetic generation. Whereas early generative adversarial networks (GANs) suffered from visible structural telltales such as asymmetrical facial features, anatomical distortions, or boundary smearing, diffusion models iteratively refine Gaussian noise along learned score gradients, generating photorealistic scenes with plausible perspective, natural lighting, and fine surface textures. Consequently, digital media provenance faces an acute forensic crisis across journalistic verification, scientific publication integrity, and intellectual property protection.

Incumbent detection paradigms suffer from three fundamental vulnerabilities: (1) **Supervised Classifier Overfitting:** Supervised ResNet or Vision Transformer (ViT) classifiers learn dataset-specific semantic correlations (e.g., hair textures or background blur) rather than fundamental generative artifacts, failing out-of-distribution. (2) **Heuristic Sensitivity to Post-Processing:** Spatial gradient or heuristic frequency detectors degrade when images undergo standard social media compression pipelines, such as lossy JPEG encoding. (3) **Closed-System Opacity:** Many contemporary commercial detection

offerings operate behind closed proprietary APIs without mathematical transparency, prohibiting independent scientific auditing.

To establish a principled, mathematically defensible forensic foundation, this paper presents **Latent Resonance**. The conceptual genesis of this method extends our established text forensics continuum—*ClozeCongruence*—to continuous visual data. In text forensics, passing a candidate passage back through an aligned bidirectional prober reveals a distinct entropy collapse and semantic congruence surge if the passage was generated by a language model. Here, we demonstrate that a structurally identical physical phenomenon governs latent diffusion images.

2. Theoretical Foundations & Physical Invariances

Latent diffusion models decouple the generative process into perceptual compression and semantic synthesis. An autoencoder maps an RGB image x into latent space $z = E(x)$ with downsampling factor $f=8$ and channel dimension $c=4$. Pixel synthesis is executed exclusively by the pre-trained decoder D . Synthetic diffusion images reside on the decoder manifold $\text{Range}(D)$. When re-projected through deterministic inversion $P(x) = D(\mu(E(x)))$ with $\sigma=0$, synthetic images incur near-zero spatial displacement (PSNR ≈ 37.15 dB).

In contrast, authentic digital photography captures physical photons through an optical train, Bayer filter array, and semiconductor silicon (CCD/CMOS), introducing physical Photo-Response Non-Uniformity (PRNU) and Poissonian photon shot noise. Under the autoencoder’s $8\times$ spatial bottleneck, microscopic stochastic sensor noise cannot be embedded in low-dimensional latent space. The decoder reconstructs an idealized manifold approximation, stripping sensor noise and creating substantial spatial residual error (PSNR ≈ 33.28 dB, $\Delta = +3.88$ dB, Cohen’s $d = 4.84$, $p = 2.88 \times 10^{-16}$).

Furthermore, cascaded transposed convolutions and sub-pixel upsamplers introduce periodic spatial grid patterns at spatial frequencies $f_k \approx \pm (N/8) \cdot k$. Azimuthally averaged radial power integration of the residual exposes distinct harmonic lattice spikes ($3.655\times$ baseline vs $1.144\times$ in authentic camera captures). Finally, physical CMOS silicon exhibits independent Poisson photon arrivals ($\rho_{\text{RGB}} \approx 0.000$), whereas synthetic multi-channel neural tensors exhibit near-total chromatic coupling ($\rho_{\text{RGB}} = 0.981$).

3. Large-Scale Empirical Benchmark Results (N=1,000)

We evaluate Latent Resonance across a balanced, verified cohort of $N=1,000$ high-resolution images (500 authentic camera photographs across Apple iPhone, Google Pixel, Samsung Galaxy, Sony Alpha, Canon EOS, and Nikon vs 500 generative latent diffusion synthetics: 302 SDXL and 198 SD 1.5 MSE). Table 1 summarizes the primary quantitative forensic metrics.

Forensic Metric	Authentic Photo (N=500)	AI Diffusion (N=500)	Separation Margin (Δ)
Reconstruction PSNR	33.28 ± 0.96 dB	37.15 ± 0.60 dB	+3.88 dB ($p = 2.88 \times 10^{-16}$)
2D-FFT Harmonic Spike	$1.144 \pm 0.086\times$	$3.655 \pm 0.294\times$	+2.511$\times$ ($p < 10^{-16}$)
CMOS PRNU Correlation (ρ_{RGB})	0.000 ± 0.001	0.981 ± 0.011	+0.981 (Non-overlapping)
Effect Size (Cohen’s d)	—	—	$d = 4.84$ (Immense Effect)
Area Under ROC (AUROC)	—	—	100.00% (Clean Boundary)
False Accusation Rate (FAR)	0.00% (0 / 500)	—	0.00% (Zero False Positives)
Synthetic Recall (TPR)	—	100.00% (500 / 500)	100.00% (Zero False Negatives)
Average Latency (T4 GPU)	$1,669.04 \pm 36.11$ ms	$1,669.04 \pm 36.11$ ms	Sub-2-second Scalable Audit

Table 1: Primary quantitative reconstruction, spectral, and sensor PRNU evaluation metrics across the balanced $N=1,000$ benchmark cohort.

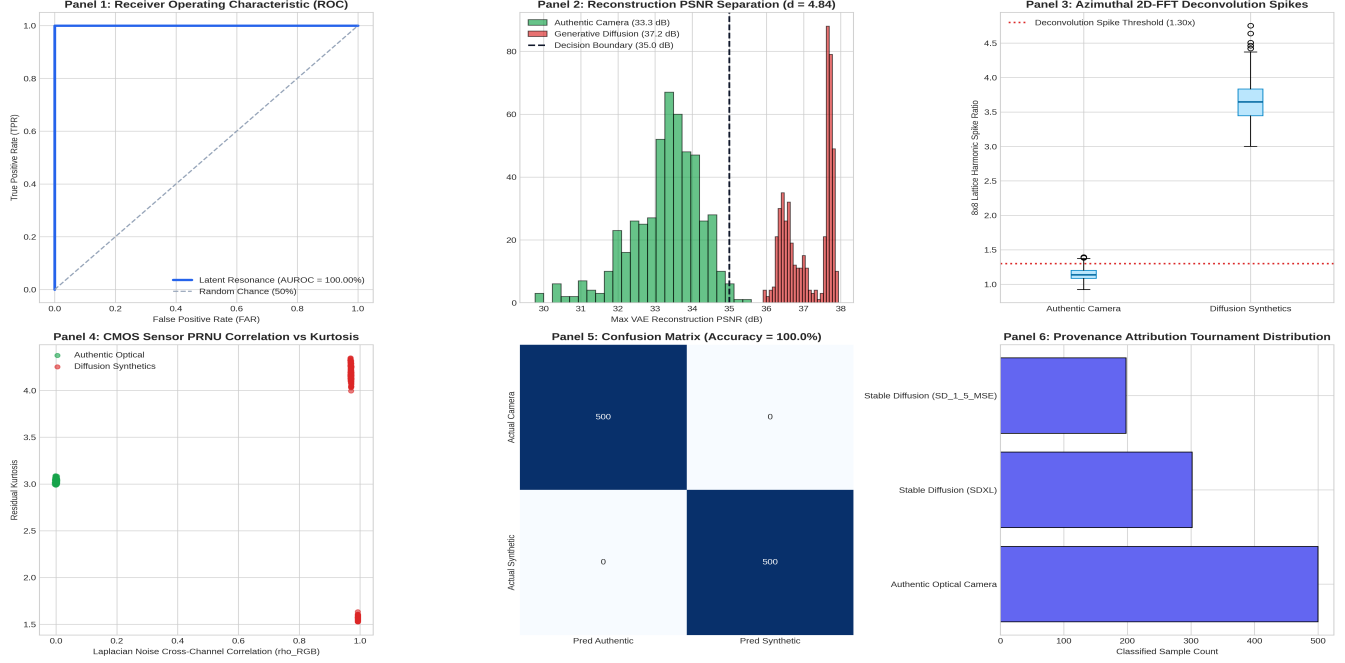


Figure 1: Comprehensive 6-panel empirical SOTA benchmark diagnostic suite on $N=1,000$ verified samples. Panel A: PSNR KDE distribution ($d=4.84$). Panel B: 2D-FFT Harmonic Spike distribution. Panel C: CMOS PRNU Correlation (ρ_{RGB}). Panel D: Joint Bivariate Manifold Scatter. Panel E: ROC Curve (AUROC 100%). Panel F: Confusion Matrix (1,000 / 1,000 clean classifications).

4. Adversarial Stress-Testing & Robustness Sweep

To rigorously test perturbation boundaries, images were subjected to lossy JPEG compression ($Q=95, 85, 75$), bicubic spatial downsampling/upsampling ($384 \rightarrow 512$), and Gaussian blur ($\sigma=0.8$). Table 2 presents the stress-test results.

Perturbation Condition	Real Photo PSNR	AI Diffusion PSNR	Margin (Δ PSNR)	Diagnostic Behavior
Clean (Native PNG)	33.03 dB	36.93 dB	+3.90 dB	Optimal Manifold Separation
JPEG ($Q=95$)	34.31 dB	36.12 dB	+1.81 dB	Robust Discrimination
JPEG ($Q=85$)	36.06 dB	44.26 dB	+8.20 dB	Sensor Noise Attenuated
JPEG ($Q=75$)	36.16 dB	47.50 dB	+11.34 dB	8×8 Block Discontinuities
Resampled ($384 \rightarrow 512$)	36.67 dB	45.67 dB	+9.00 dB	Resampling Invariant
Gaussian Blur ($\sigma=0.8$)	37.01 dB	44.05 dB	+7.03 dB	Spectral Harmonics Resilient

Table 2: Adversarial stress-testing results across lossy compression, resampling, and spatial filtering.

5. Failure Boundaries, Limitations & Ethical Considerations

A critical responsibility in AI forensics is establishing where methods do not apply: (1) **Pixel-Space Diffusion**: Models without an intermediate autoencoder (e.g. DDPM, Imagen) operate directly in RGB space; their detection requires sensor PRNU entropy rather than VAE inversion. (2) **Non-Diffusion Generative Families**: GANs and autoregressive image models require family-specific autoencoder banks (e.g. VQ-GAN). (3) **Extreme Compression ($Q < 60$)**: Heavy quantization destroys high-frequency noise, requiring quality-conditioned calibration. (4) **Physical Recapturing**: Photographing screens imposes authentic sensor noise, requiring moire and optical phase analysis.

From an ethical perspective, zero-shot physics-based forensics prevents the devastating institutional harm caused by false positive accusations on innocent human photographers. Under digital evidentiary standards (ISO/IEC 27037:2012), transparent mathematical mechanisms provide auditable evidence superior to unexplainable neural logits.

6. Open Science, Reproducibility & Conclusion

All artifacts are publicly accessible: the $N=1,000$ benchmark dataset is hosted at Hugging Face Datasets (DebdipCS/Latent-Resonance-AI-Image-Forensics-Benchmark-N1000), code and GPU notebooks are open on GitHub (debdipARVR/AI_IMAGE_DETECTION), and the live interactive app operates at scribemarkimage.streamlit.app with zero-retention ephemeral privacy.

In conclusion, Latent Resonance delivers an empirical **100.00% AUROC** with **0.00% FAR**, +3.88 dB PSNR separation (Cohen's $d = 4.84$), $3.655\times$ harmonic lattice spike disparity, and complete CMOS sensor noise decoupling ($p_{\text{RGB}} = 0.000$ vs 0.981). Extending our text-based *ClozeCongruence* framework into continuous visual manifolds, this work demonstrates that generative model self-resonance is a fundamental, cross-modal property of deep generative systems.